First-order Stochastic Algorithms for Escaping From Saddle Points in Almost Linear Time

Yi Xu, Rong Jin and Tianbao Yang

NeurIPS 2018

List item 49 / Ref 76 | Tags: S, E, G / F, B | Tier: Must read

The paper in one minute

Question: Can stochastic first-order gradients reveal negative curvature with nearly linear cost in parameter dimension?
Model/Method: NEON/NEON+: a specially centered gradient iteration amplifies directions where the Hessian is negative, without explicitly computing Hessian-vector products.
Finding: NEON provides provable negative-curvature directions; combining it with first-order stochastic optimization yields high-probability second-order stationarity, including a stated $\widetilde{O}(d/\varepsilon^{3.5})$ runtime regime.
Why we care: The Hessian's escaping directions can be detected indirectly using the same primitive as backpropagation: gradients.

1. Why first-order information can reveal second-order geometry

A stochastic nonconvex objective has the form $F(x) = \mathbb{E}_\xi[f(x;\xi)]$. We might know only sampled gradients $\nabla f(x;\xi)$, not the full Hessian. At a stationary saddle, $\nabla F(x)$ is nearly zero while $\lambda_{\min}(\nabla^2 F(x)) < 0$. NEON asks whether we can discover this negative direction using *gradient differences*.

Start with a centered auxiliary function

Define, at fixed reference point x,

$$\widehat{F}_x(u) = F(x+u) - F(x) - \nabla F(x)^T u.$$

By Taylor's theorem, when u is small,

$$\widehat{F}_x(u) \approx \tfrac{1}{2} u^T H u, \qquad \nabla \widehat{F}_x(u) = \nabla F(x+u) - \nabla F(x) \approx Hu,$$

where $H = \nabla^2 F(x)$. Notice: *two gradients give a local approximation to a Hessian-vector product*. GD on this centered function therefore behaves near the origin as

$$u_{t+1} \approx (I - \eta H) u_t.$$

A negative Hessian eigenvalue $-\gamma$ becomes multiplier $1 + \eta\gamma > 1$, so its component grows faster than directions of positive curvature.

Worked eigenvalue example

Let $H = \mathrm{diag}(2, -1)$, $u_0 = (0.1, 0.1)$ and $\eta = 0.2$. Then

$$u_1 = (0.6 \cdot 0.1,\ 1.2 \cdot 0.1) = (0.06, 0.12),$$

$$u_2 = (0.036, 0.144).$$

The positive-curvature component shrinks while the negative one grows. After normalization, u_t becomes increasingly aligned with the escaping eigenvector $(0, 1)$. This is the NEON intuition, not the paper's full stopping or confidence argument.

2. From intuition to NEON and NEON+

NEON (Algorithm 1 and Theorem 1)

Start with a small random u_0 around the reference point x. Repeatedly apply gradient steps to the locally centered objective and monitor a specified decrease test. If the objective exhibits sufficient negative curvature, the iteration can find a vector u with

$$\frac{u^T \nabla^2 F(x) u}{\|u\|^2} \leq -\widetilde{\Omega}(\gamma)$$

with high probability when $\lambda_{\min}(\nabla^2 F(x)) \leq -\gamma$, subject to the theorem's parameter conditions. If it returns the designated null result, it provides a corresponding negative-curvature absence conclusion to tolerance and confidence.

Why NEON+ matters

Ordinary gradient iterations grow the negative-curvature component on a scale related to $1/\gamma$. NEON+ accelerates the centered dynamics using a momentum-style/Nesterov mechanism, improving the negative-curvature search's dependence toward $1/\sqrt{\gamma}$ (up to logarithmic and problem factors). This is important because a small negative eigenvalue is otherwise hard to detect.

Use the escaping vector to descend

If v is normalized and satisfies $v^T H v \leq -c\gamma$, choose an appropriate signed step of scale γ/L_2 along v, where L_2 is Hessian Lipschitz constant. A Taylor bound implies expected or deterministic objective decrease of order γ^3/L_2^2 (Lemma 1). Thus the algorithm converts *detected negative curvature* into genuine improvement.

Framework and rates

The paper combines NEON-type curvature search with first-order optimizers (SGD, stochastic momentum, SCSG). In its stated stochastic setting and runtime model, a NEON–SCSG variant finds an $(\varepsilon, \sqrt{\varepsilon})$-approximate second-order stationary point with high probability in $\widetilde{O}(d/\varepsilon^{3.5})$ time. The *almost linear* claim concerns dependence on d and the chosen oracle-cost model; it is not an $O(\varepsilon^{-3.5})$ convergence rate without any dimensional or regularity factors.

> **What to read:** Section 3's assumptions, Section 4's centered function / Algorithms 1–2 / Theorem 1, Section 5's framework and Table 1, then Figure 1 and experiments in Section 6. Appendix is needed only for proof constants.

3. What the authors establish and where it stops

Contributions. (i) A general first-order way to detect Hessian negative curvature using gradient-difference dynamics; (ii) an accelerated NEON+ variant; (iii) combinations with stochastic first-order optimizers providing second-order guarantees; (iv) numerical comparisons with power and Lanczos methods in selected models. **Assumptions.** Individual sample objectives require smooth gradients and Hessians, and the stochastic-gradient noise obeys a stated exponential-moment/bounded-variance-type condition (Assumption 1). The global complexity also assumes a finite initial objective gap. Parameter choices depend on smoothness constants and target curvature threshold. **Limitations.** These are oracle/runtime guarantees for smooth stochastic objectives, not proofs of global minima for arbitrary deep ReLU networks. Explicit randomization is still used to seed the negative-curvature search. Claimed speedups depend on cost models and dimension; full practical deep-learning speed is not proven.

Important comparisons

Jin (2017) injects perturbations to induce escape directly. **Daneshmand (2018)** assumes intrinsic stochastic gradients align with negative curvature (CNC). Xu–Jin–Yang instead *constructs* a gradient-only negative-curvature search mechanism. **NEON2 (Ref 77)** replaces/strengthens this detection primitive with a different reduction, improving several oracle complexities.

Quick reference

Quantity	Interpretation
$\widehat{F}_x(u)$	Centered local function; removes constant and linear terms.
$\nabla F(x+u) - \nabla F(x)$	First-order proxy for Hu.
$(I - \eta H)u$	Linearized iteration; negative eigendirections expand.
$u^T Hu/\|u\|^2 < 0$	Certificate of a negative-curvature direction.
$\widetilde{O}(d/\varepsilon^{3.5})$	Particular stochastic high-probability runtime, not a universal GD bound.

Papers Master note

Xu–Jin–Yang introduce NEON and NEON+, gradient-only negative-curvature detectors based on centered gradient differences. Coupled with stochastic first-order methods they obtain high-probability approximate second-order stationarity, including nearly linear-in-dimension runtimes under smoothness and stochastic-noise assumptions.

Source: original NeurIPS 2018, Sections 3–6, Theorem 1, Table 1.